

Rafael Antonio Casella

Financial Data Scientist

+55 (17) 99763-5500 rafacasella@gmail.com rafacasella.github.io São José do Rio Preto, SP, Brazil

Professional Summary

Analytical and results-driven Data Scientist with a solid background in financial markets and quantitative analysis. Focused on process automation using Python, SQL querying, and building strategic dashboards in Power BI. Proven track record in developing predictive models, machine learning architecture, and end-to-end data pipelines to drive financial performance and strategic decision-making.

Professional Experience

Financial and Data Specialist 09/2025 – 05/2026
Ceopag (Payment Fintech) São José do Rio Preto, SP, Brazil

- Actively supported strategic financial data analysis projects, extracting key insights for executive-level reports.
- Executed complex bank reconciliations and managed accounts payable/receivable workflows, ensuring data integrity for auditing purposes.

Market Analyst 02/2017 – 07/2025
Self-Employed B3 Trader (BM&F-BOVESPA) Brazil

- Executed structured operations in USD and Index Future Contracts using quantitative models.
- Managed capital risk through agile decision-making based on intraday statistical patterns and technical market analysis.

Co-Founder & Technical Analyst (CNPI-T) 06/2018 – 12/2021
Portal do Mercado São Paulo, SP, Brazil

- Led the technical management of the NysePro software, structuring real-time news streaming pipelines and metric consolidation.
- Developed and optimized recommended investment portfolios based on quantitative data analysis for the community members.

Education

Postgraduate Degree in Data Science, Machine Learning & AI
XP Educação (In Progress)

Bachelor’s Degree in Computer Engineering
UFSCar / UNORP

Certifications

- Python for Data Analysis and Machine Learning (64h) — Data Science Academy
- Advanced Power BI & Excel (48h) — Tetra Academy

Languages

Portuguese: Native | English: Advanced / Professional Full Proficiency

Technical Skills

Languages & Frameworks

Python Pandas Numpy Matplotlib

Databases & SQL

PostgreSQL MySQL

Machine Learning & Deep Learning

Scikit-Learn Statsmodels TensorFlow Keras

Automation & Visualization

Power BI Seaborn Streamlit Playwright

Key Projects

Customer Churn Predictor

Python, Scikit-Learn, Logistic Regression

- Addressed high monthly churn rates by building and deploying a Logistic Regression model to identify at-risk customers.
- Achieved **82.15% accuracy**, providing early alerts that enabled proactive retention operations.

Quantitative Trading Pipeline & Analytical Dashboard

Python, Streamlit, Machine Learning, Backtesting

- Engineered an end-to-end infrastructure for modeling quantitative financial algorithms, completely mitigating look-ahead bias.
- Developed a modular ETL pipeline and applied feature engineering to extract momentum and volatility indicators.
- Deployed an analytical Streamlit dashboard to monitor critical portfolio KPIs, including Sharpe Ratio and Maximum Drawdown.

Geopolitical Intelligence Monitor – Truth Social

Python, Playwright, CustomTkinter, Multithreading

- Developed an automated, real-time pipeline to monitor strategic keywords impacting global financial markets.
- Implemented a multithreading structure combined with Asyncio to isolate the browser worker, preventing GUI freezing.
- Replaced traditional web scraping with network request interception, capturing raw JSON payloads directly from the API.

Stock Price Forecasting using LSTM

TensorFlow, Keras, MinMaxScaler, RNN

- Overcame the limitations of traditional statistical models in capturing complex temporal dependencies in financial time series.
- Implemented a Long Short-Term Memory (LSTM) Recurrent Neural Network with Dropout layers to prevent overfitting.
- Applied feature scaling using MinMaxScaler and validated predictive performance using RMSE and MAE metrics.